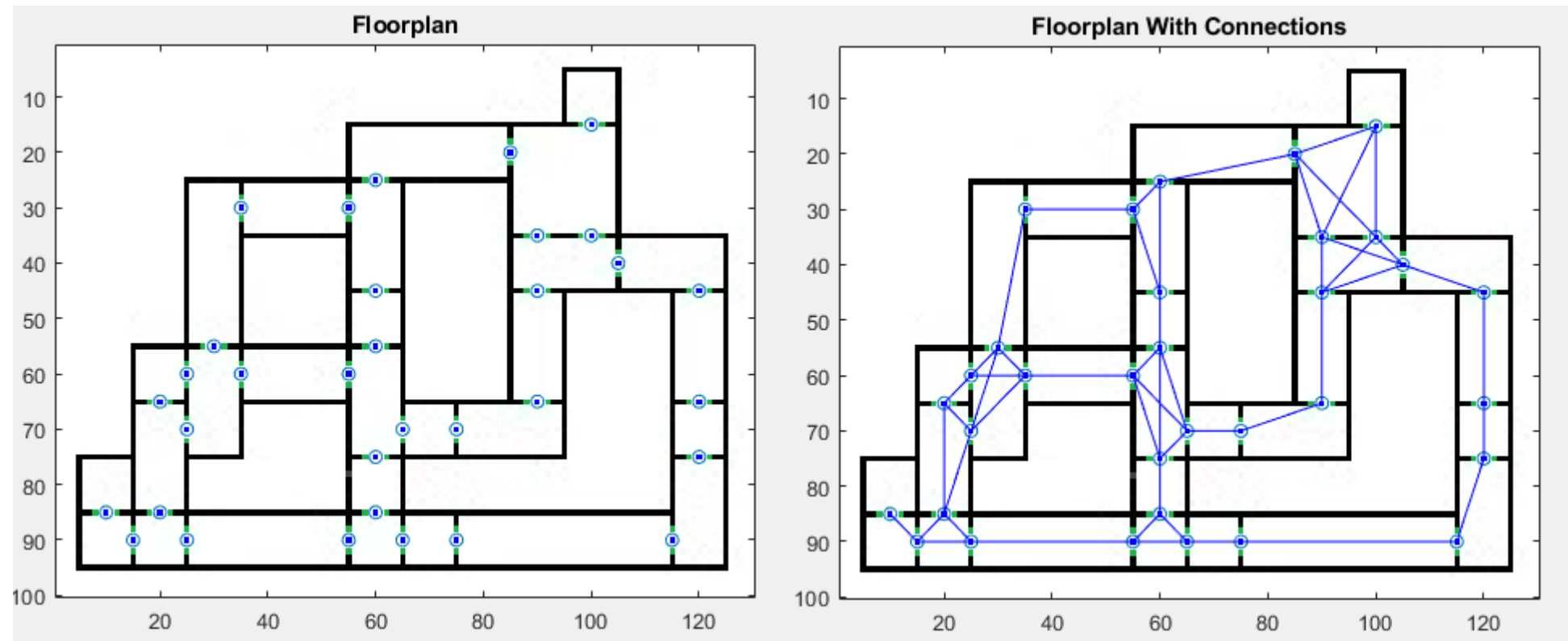


A* High Level Planning & Replanning

A* Setup

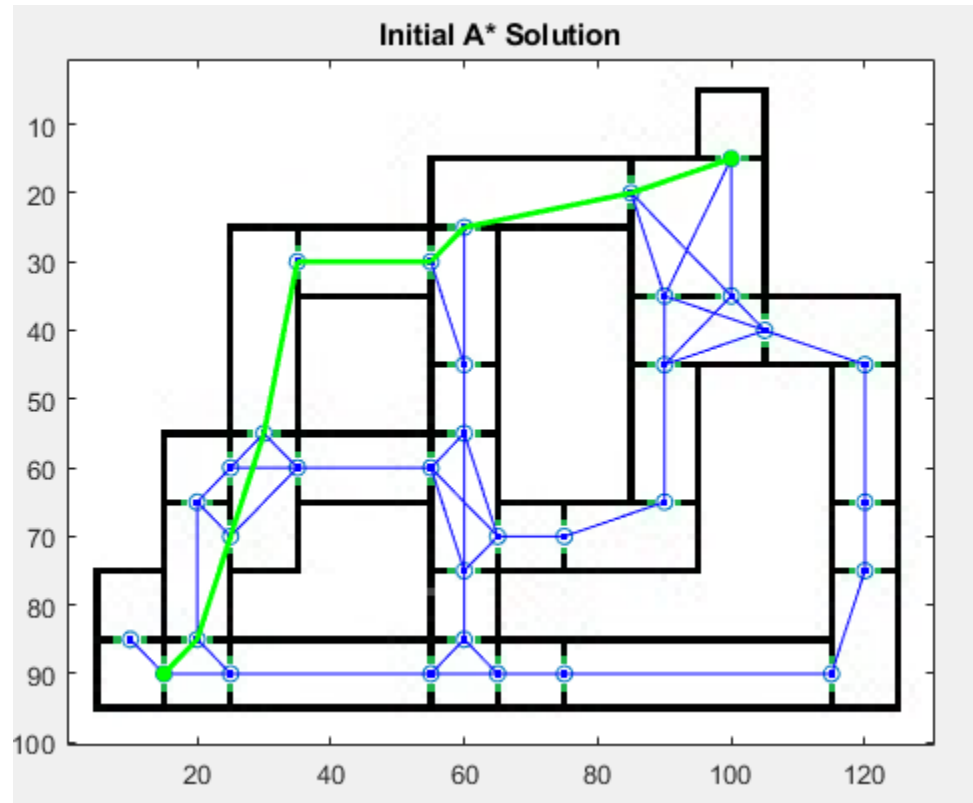
In a Known environment a tree of the portals and their connections can be created such that an A* algorithm can solve the ideal order of rooms/portals to travel from one room to another



Example Floorplan with multiple possible paths to any one destination

A* Solution

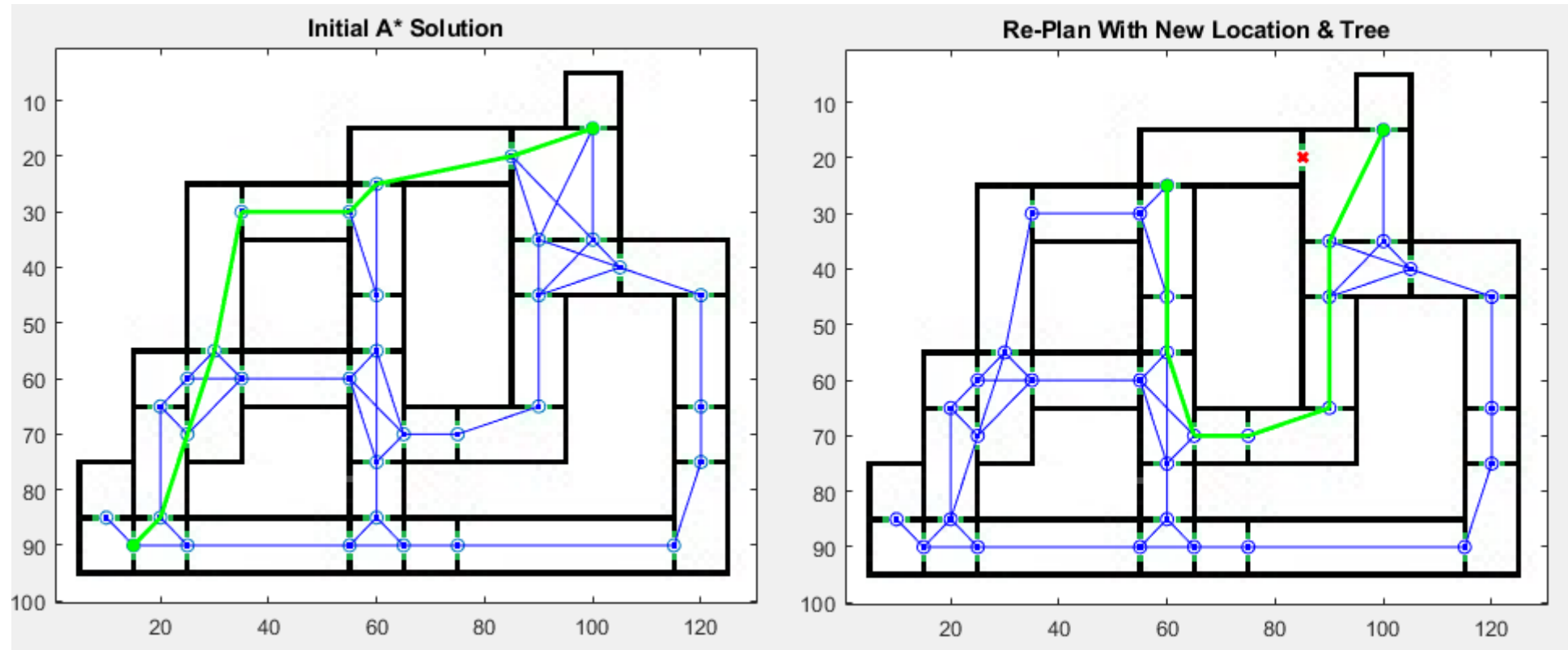
Feeding the tree of connections to an A* algorithm along with start and end points a solution is determined



The initial A* solution starting in the bottom left corner traveling to the top right

A* Re-Plan

If in the course of traveling along this path we determine that a portal we need to use is blocked we can remove this portal and its connections from the tree and call the A* script once more to achieve a new path from the current location



Initial solution alongside new solution once we have traveled to the room with the previously unknown blockage and determined the need to replan